

EXHIBIT 22

Ex. 22 - Infringement Contention Claim Chart
U.S. Patent No. 12,332,352 (“the ’352 patent”)

Apple’s products directly and/or indirectly infringe at least claims 1, 2, 9, 14, 16, 23, 28 (“Asserted Claims”) of U.S. Patent No. 12,332,352 (“the ’352 patent”). The products that infringe the ’352 patent include at least the iPhone 15, iPhone 15 Plus, iPhone 15 Pro, iPhone 15 Pro Max, iPhone 16, iPhone 16e, iPhone 16 Pro, iPhone 16 Pro Max, iPhone 16 Plus, iPhone 17, iPhone 17e, iPhone 17 Pro, iPhone 17 Pro Max, iPhone Air, 11-inch iPad Pro (4th Generation), iPad Pro 12.9 inch (6th Generation), iPad Pro 11-inch (M4 and M5), iPad Pro 13-inch (M4 and M5) and other infringing iPhone and iPad models (each an “Accused Apple Product” and collectively the “Accused Apple Products”). The Accused Apple Products infringe the ’352 patent as set forth below. Each of the following limitations is met literally, and, to the extent a limitation is not met literally, it is met under the doctrine of equivalents.

Additionally, the Apple iPhone 17 is representative of the Accused Apple Products at least because the other Accused Apple Products function in the same or substantially similar manner to the Apple iPhone 17.¹ The relevant identification processes and functionalities, as well as the hardware components of the Apple iPhone 17, are demonstrated through testing of the Apple iPhone 17 (“Apple Product Testing”), which is available for inspection upon request.

Claim 1	Accused Apple Products
[1pre] A detector for identifying at least one material property m comprising	The Accused Apple Product comprise a detector for identifying at least one material property m. For example, the Apple iPhone 17 includes a TrueDepth camera system that enables features like FaceID

¹ Each of the Accused Apple Products includes the claimed hardware components and steps. The hardware components and source code implementing the relevant functionalities of the Apple iPhone 17 are representative of the Accused Apple Products for the purpose of infringement. To the extent there may be differences in the hardware composition, certain software steps, or source code versions among these Accused Apple Products, those differences are not material for the infringement analysis.

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Exhibit 36.

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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

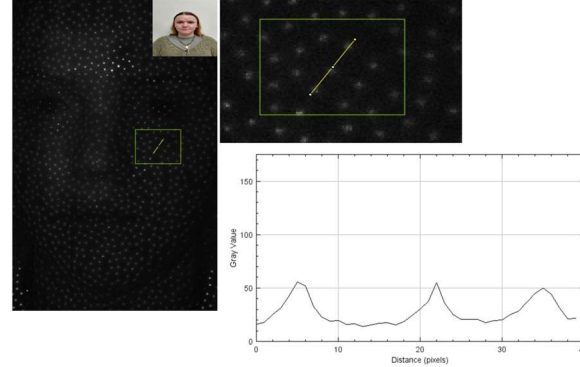
After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

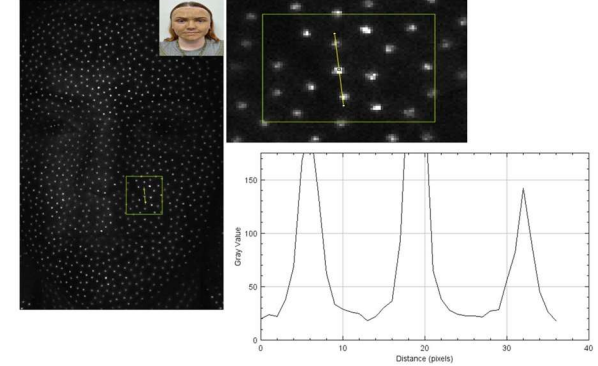
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Face ID detects material properties. Face ID on the Apple iPhone 17, for example, succeeds when analyzing certain materials, like skin, but fails when analyzing other materials, like a facial mask, despite the geometry of the analyzed face remaining consistent.

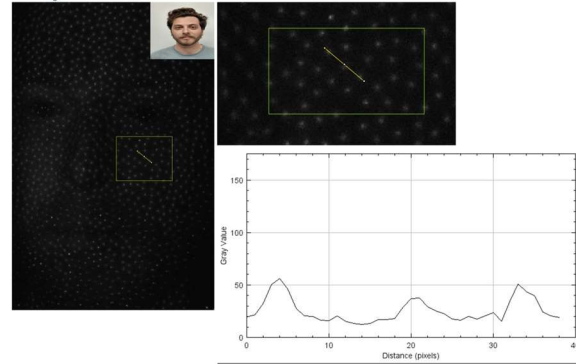
Subject01 Covered area: none – natural Face



Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face

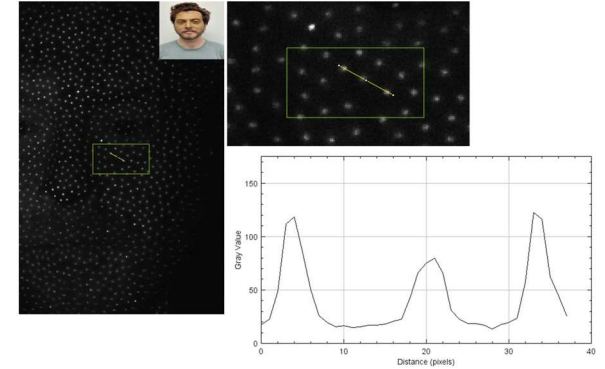


Exhibit 47 ¶¶ 23-31.

[1a] at least one light projector,
 wherein the light projector is

The Accused Apple Product comprise at least one light projector, wherein the light projector is configured for generating at least one illumination pattern for illumination of the object, wherein

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configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern; a pseudo-random point pattern; a random point pattern or a quasi-random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; and at least one line pattern comprising parallel or crossing lines;	the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern; a pseudo-random point pattern; a random point pattern or a quasi-random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; and at least one line pattern comprising parallel or crossing lines. For example, the Apple iPhone 17 includes a TrueDepth camera system that enables features like FaceID. Face ID Enabled by TrueDepth technology in the Center Stage front camera Exhibit 33.
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APPLE IPHONE 17 DISASSEMBLY



Exhibit 34C, at 23.

The TrueDepth camera system includes a dot projector Vertical-Cavity Surface-Emitting Laser (“VCSEL”), a flood illumination VCSEL, and a near-infrared camera module.

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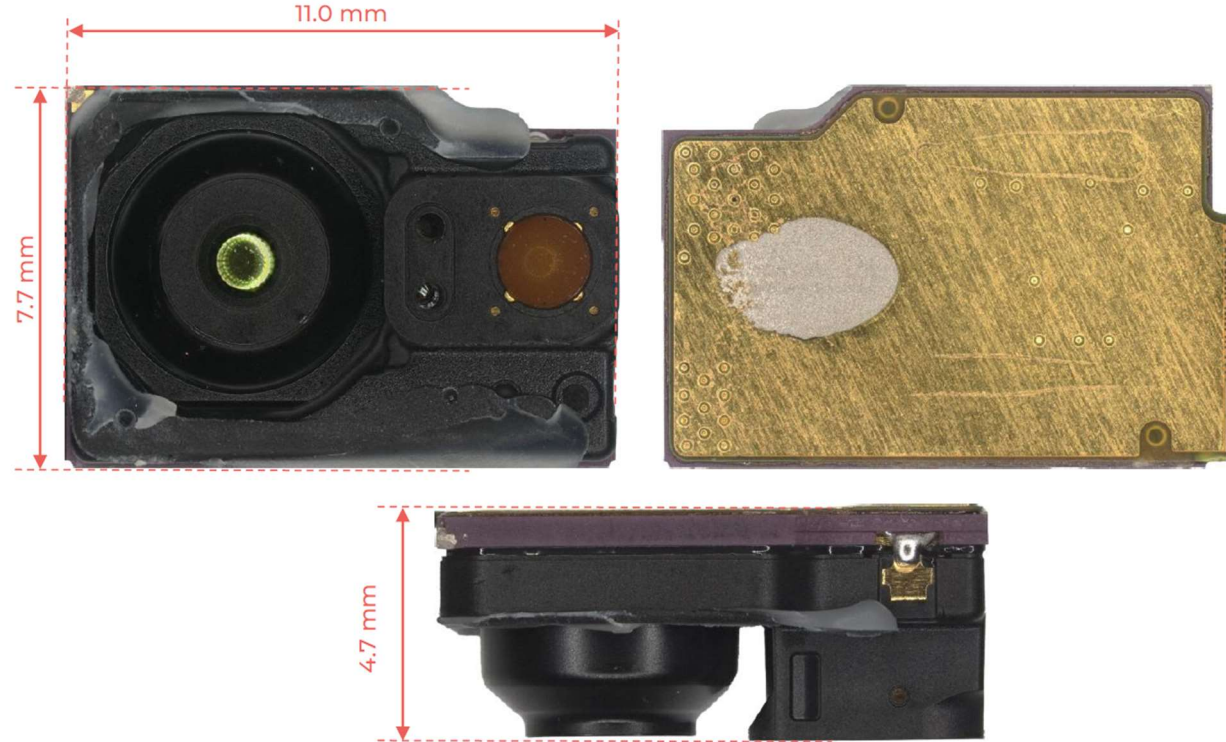


Exhibit 34C, at 35.

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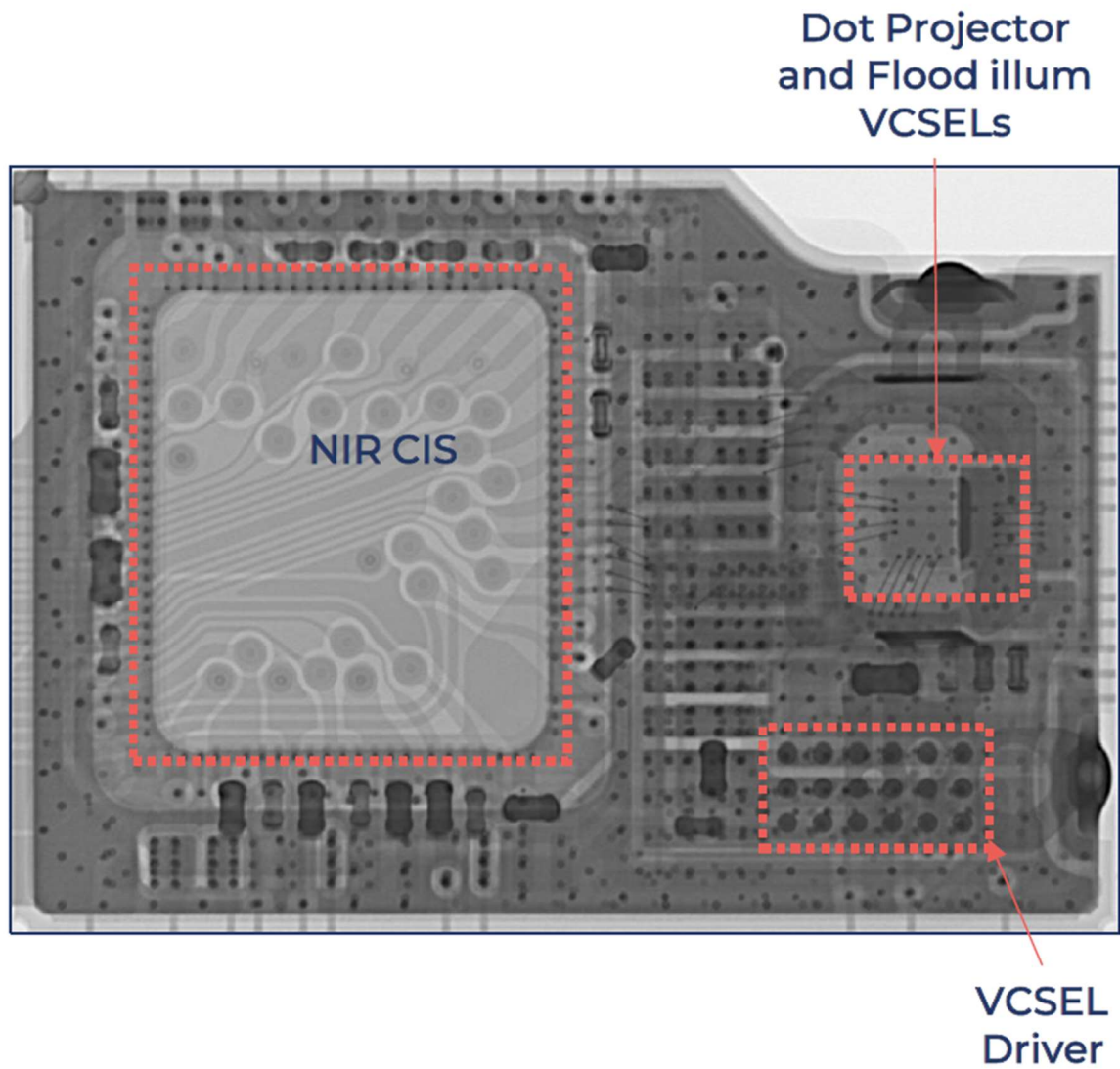


Exhibit 34C, at 36.

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iPhone 17 Pro Face ID
Module Opened © Yole
Group 2026

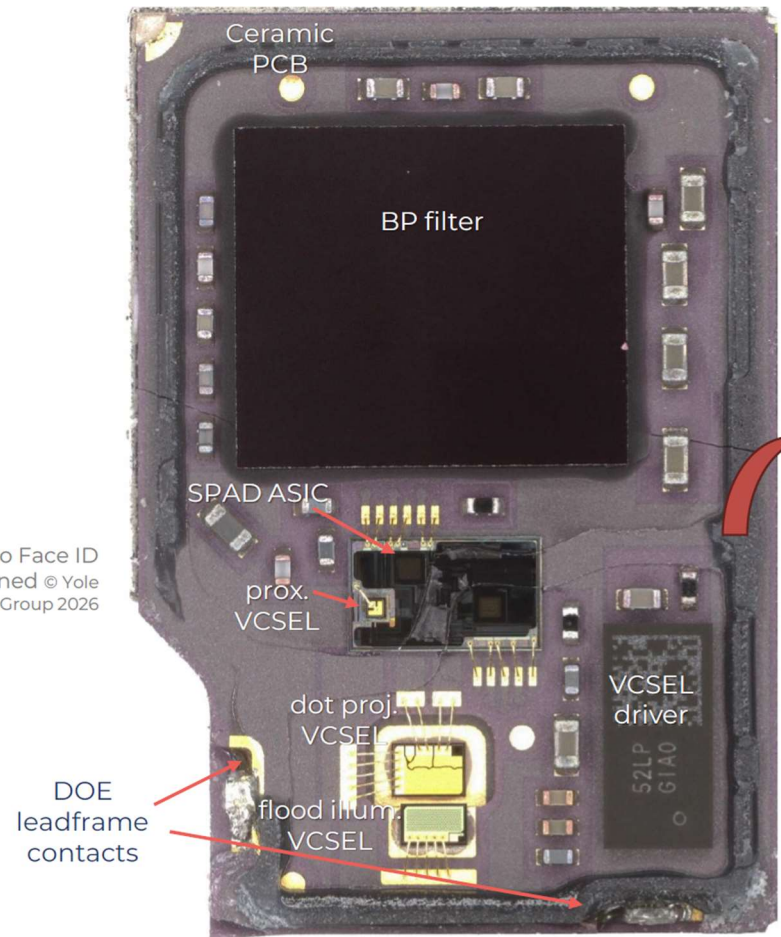


Exhibit 34C, at 37.

The TrueDepth camera system generates at least one illumination pattern for illumination of the object.

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	After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses. Exhibit 35, at 22. For example, the dot projection VCSEL employs the following illumination pattern.
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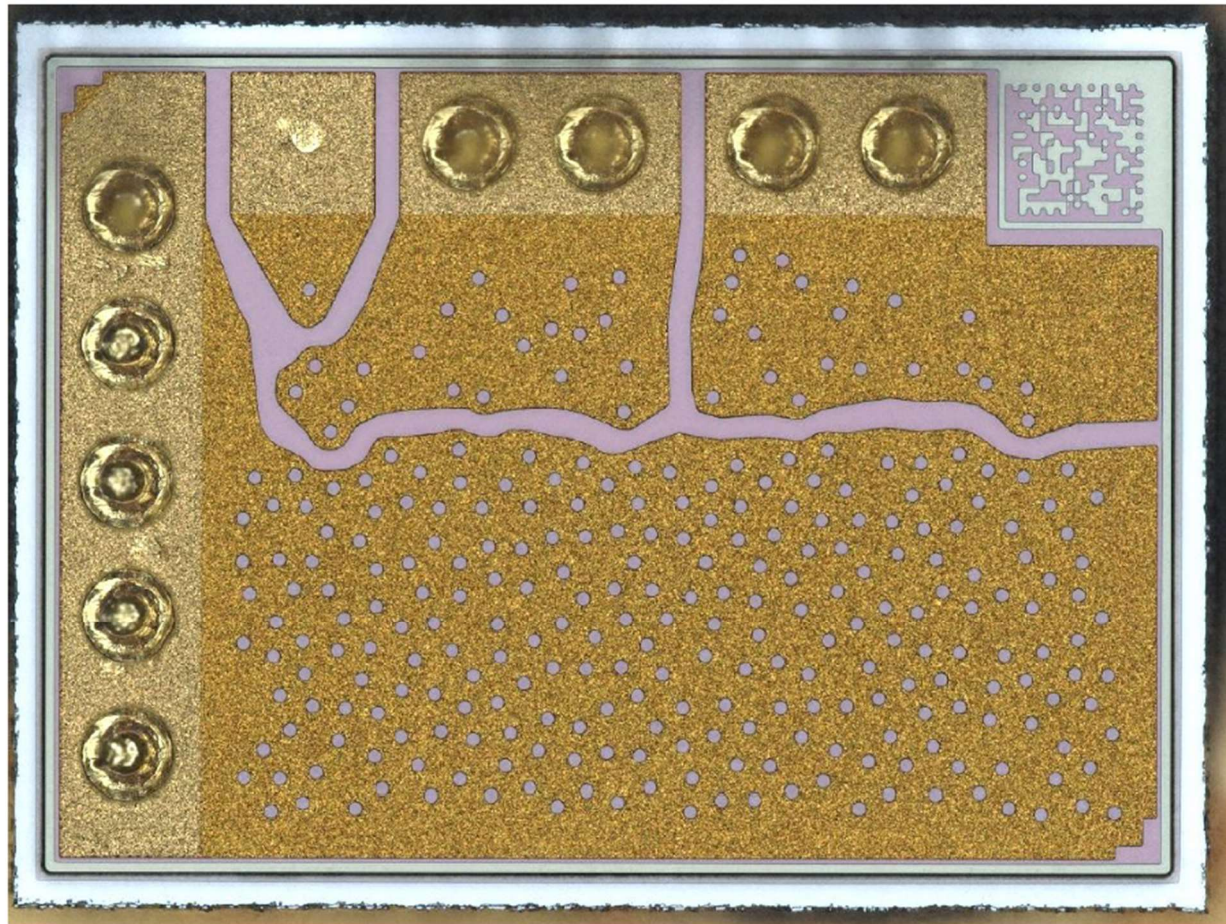
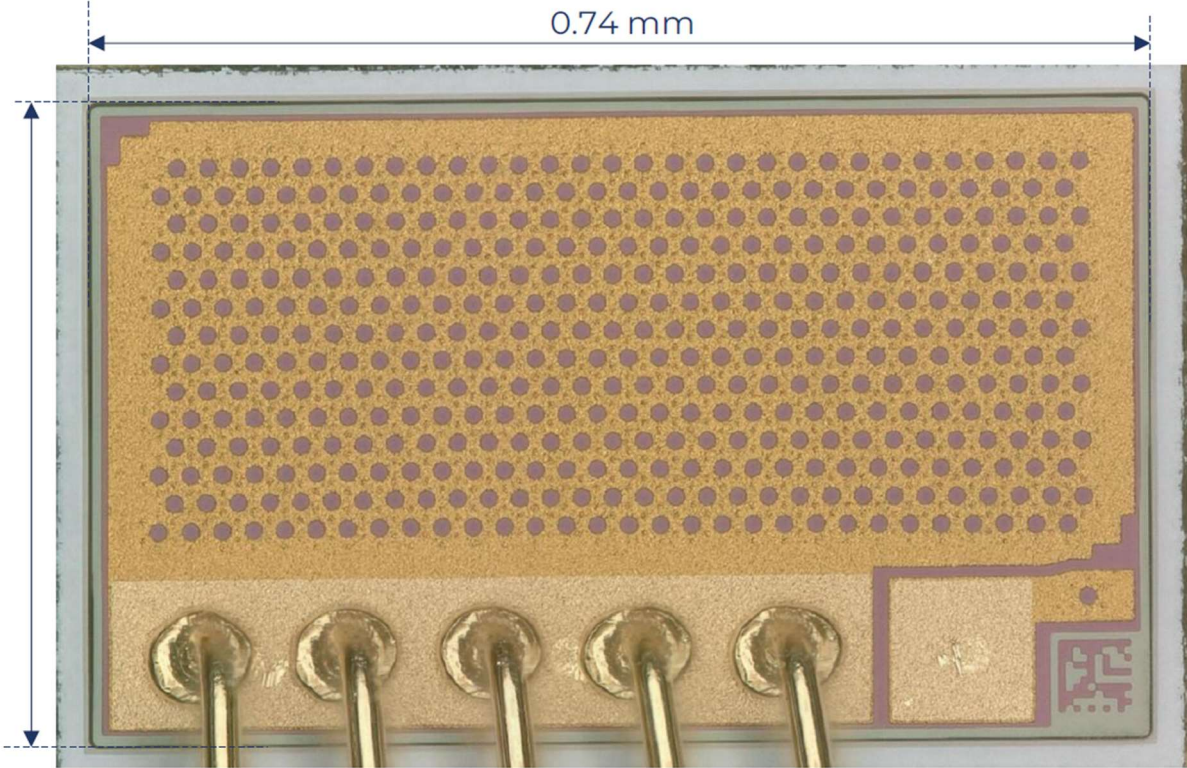


Exhibit 34C, at 77.

The flood arrangement VCSEL employs the following illumination pattern.

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	 A micrograph of a camera sensor array. The array is rectangular and contains a grid of small, circular optical sensors. The dimensions are indicated as 0.74 mm in width and 0.45 mm in height. The array is mounted on a substrate, and several gold-colored wires are visible at the bottom edge. Exhibit 34C, at 86.
[1b] at least one camera comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the camera is configured for recording at least one reflection image of a light beam originating from at least one object; and	The Accused Apple Products comprise at least one camera comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the camera is configured for recording at least one reflection image of a light beam originating from at least one object. For example, the Apple iPhone 17 includes a camera configured for recording at least one reflection image of a light beam originating from at least one object.

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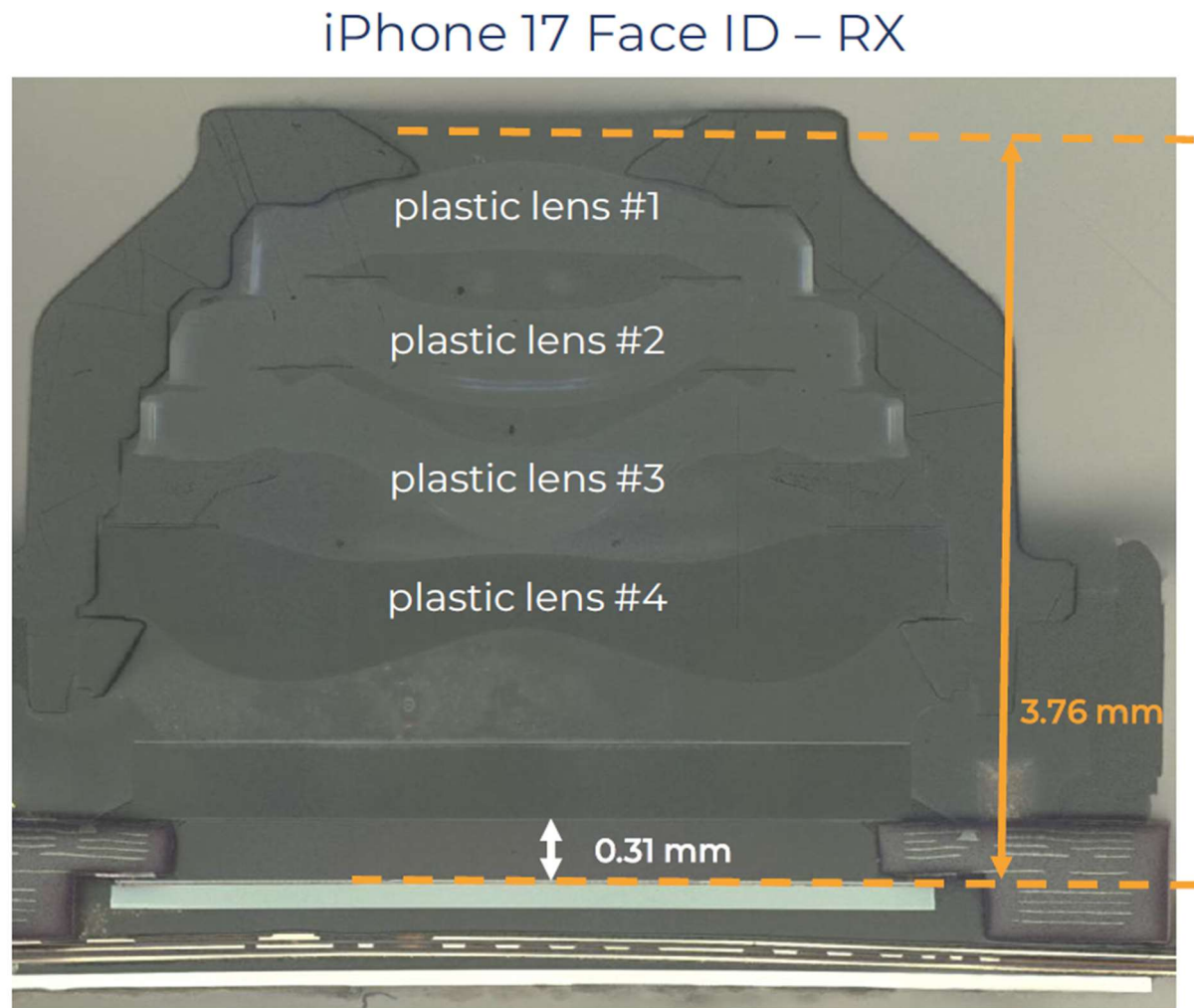


Exhibit 34C, at 27.

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The camera contains a matrix of optical sensors having light-sensitive areas. For example, a Complementary Metal Oxide Semiconductor (“CMOS”) sensor.

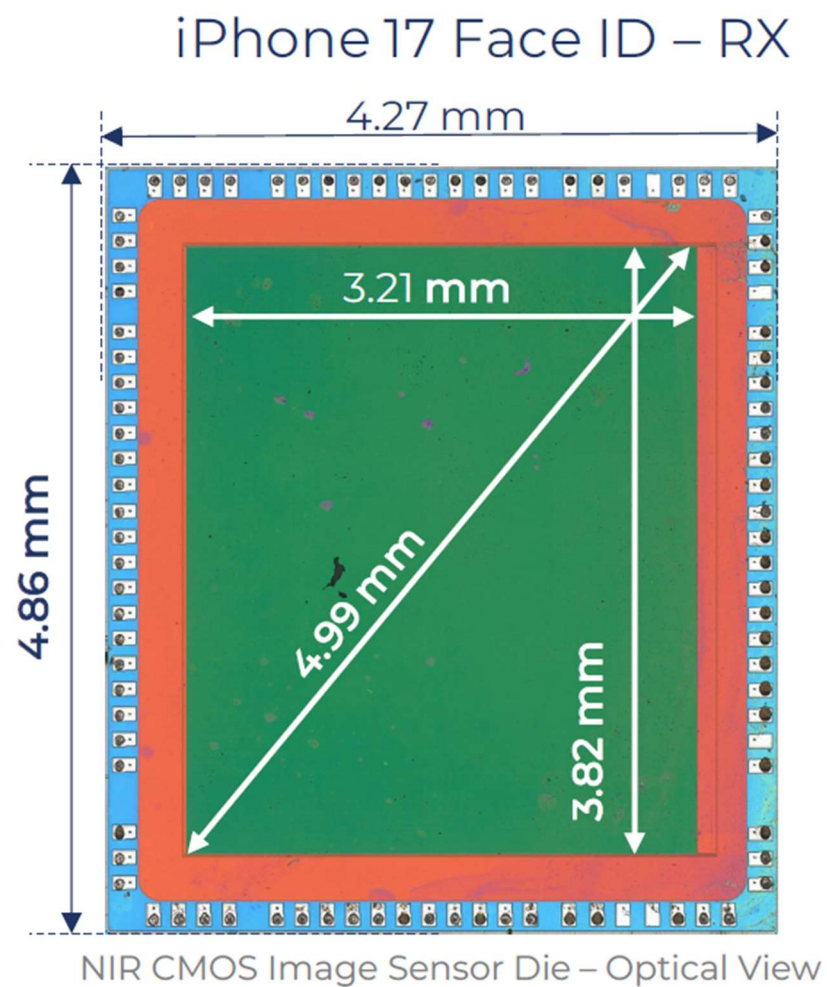


Exhibit 34C, at 28.

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	The camera records reflection images of a light beam originating from at least one object. Apple’s biometric security architecture relies on a strict separation of responsibilities between the biometric sensor and the Secure Enclave, and a secure connection between the two. The sensor captures the biometric image and securely transmits it to the Secure Enclave. During enrollment, the Secure Enclave processes, encrypts, and stores the corresponding Optic ID, Face ID, and Touch ID template data. During matching, the Secure Enclave compares incoming data from the biometric sensor against the stored templates to determine whether to unlock the device or respond that a match is valid (for Apple Pay, in-app, and other uses of Optic ID, Face ID, and Touch ID). The architecture supports devices that include both the sensor and Secure Enclave (such as iPhone, iPad, many Mac computers, and Apple Vision Pro), as well as the ability to physically separate the sensor into a peripheral that’s then securely paired to the Secure Enclave in a Mac with Apple silicon. Exhibit 35, at 20. After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses. Exhibit 35, at 22.
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	Advanced technologies The technology that enables Face ID is some of the most advanced hardware and software that we’ve ever created. The TrueDepth camera captures accurate face data by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face. A portion of your device's neural engine — protected within the Secure Enclave — transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data. Exhibit 36.
[1c] at least one processor configured for determining the material property by evaluation of at least one beam profile of the reflection image,	The Accused Apple Products comprise at least one processor configured for determining the material property by evaluation of at least one beam profile of the reflection image. For example, the Apple iPhone 17 includes a processing chip. Chip  A19 chip 6-core CPU with 2 performance and 4 efficiency cores 5-core GPU with Neural Accelerators 16-core Neural Engine Hardware-accelerated ray tracing Exhibit 33.

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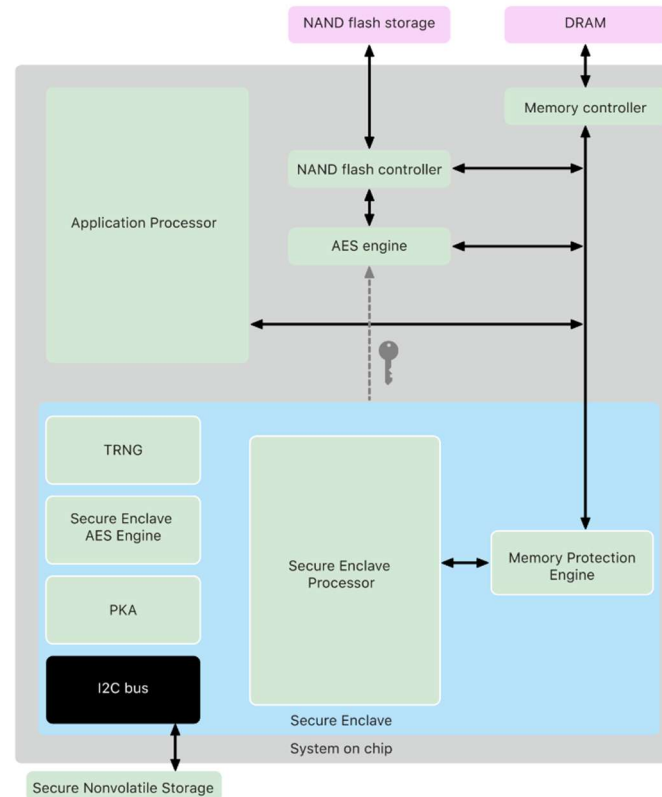
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Secure Enclave

The Secure Enclave is a dedicated secure subsystem in the latest versions of iPhone, iPad, Mac, Apple TV, Apple Watch, Apple Vision Pro, and HomePod.

Overview

The Secure Enclave is a dedicated secure subsystem integrated into Apple [system on a chip \(SoC\)](#). The Secure Enclave is isolated from the main processor to provide an extra layer of security and is designed to keep sensitive user data secure even when the Application Processor kernel becomes compromised. It follows the same design principles as the SoC does—a boot ROM to establish a hardware root of trust, an AES engine for efficient and secure cryptographic operations, and protected memory. Although the Secure Enclave doesn't include storage, it has a mechanism to store information securely on attached storage separate from the NAND flash storage that's used by the Application Processor and operating system.



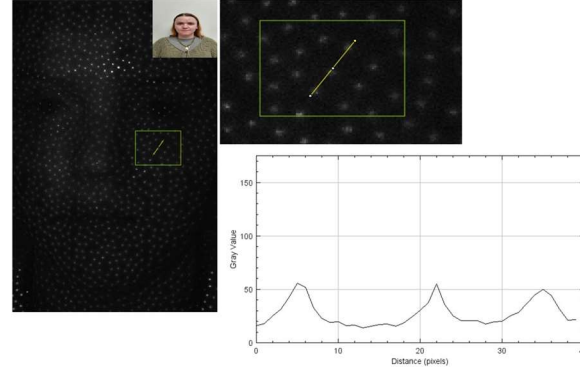
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Exhibit 35, at 10.

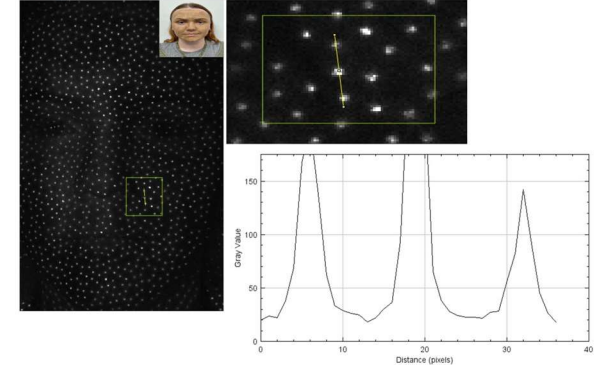
The processor is configured for determining the material property by evaluation of at least one beam profile of the reflection image.

For example, Face ID on the Apple iPhone 17 succeeds when evaluating certain beam profiles, which indicate one material, and fails when evaluating other beam profiles, which indicate a different material.

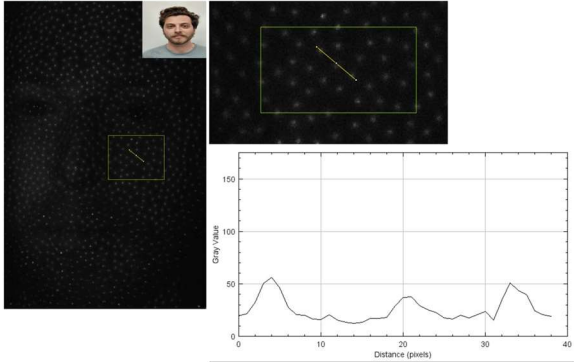
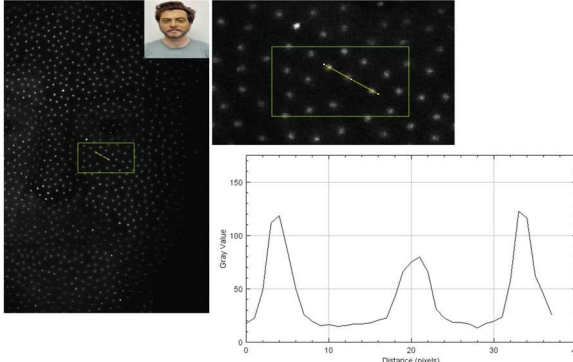
Subject01 Covered area: none – natural Face



Subject01 Covered area: entire face

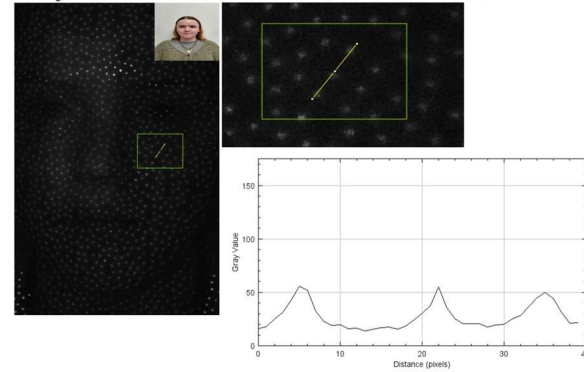


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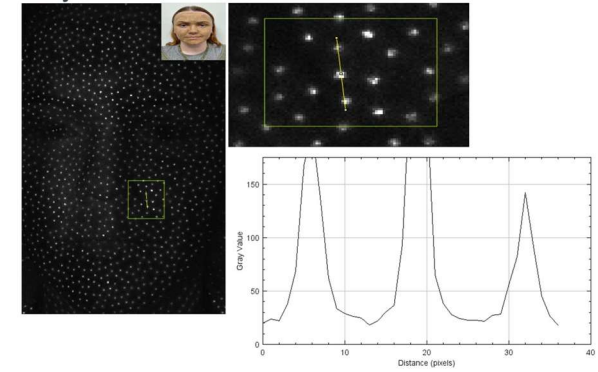
	Subject02 Covered area: none – natural Face  Subject02 Covered area: entire face  Exhibit 47 ¶¶ 23-31.
[1d] wherein the processor is configured for determining at least one material feature $\phi 2m$ by applying at least one material dependent image filter $\Phi 2$ to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; and a linear combination thereof;	The Accused Apple Products comprise a processor that is configured for determining at least one material feature $\phi 2m$ by applying at least one material dependent image filter $\Phi 2$ to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; and a linear combination thereof. For example, Face ID on the Apple iPhone 17 applies a material dependent image filter that succeeds when analyzing one material and fails when analyzing another material—characterized, for example, by differences in spot shape, luminance, contrast, energy, homogeneity, and dissimilarity—even though the geometry of the face stays the same.

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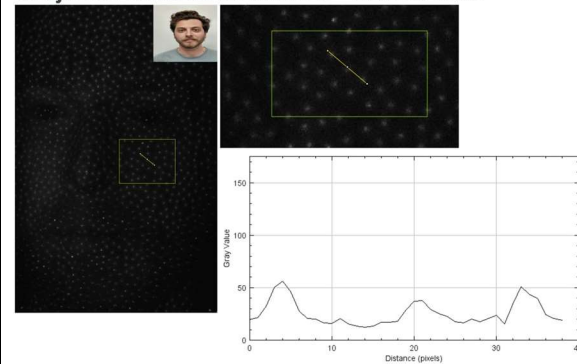
Subject01 Covered area: none – natural Face



Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face

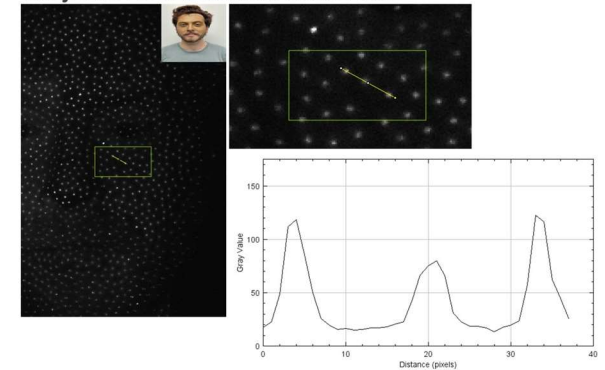


Exhibit 47 ¶¶ 23-31.

Further, Face ID uses neural networks that consider this information, provided by the given image, as well.

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	Face ID security With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data. Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing. The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask. After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses. Exhibit 35, at 22.
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Facial matching security

Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users. Face ID is designed to work with hats, scarves, eyeglasses, contact lenses, and many types of sunglasses. Face ID also supports unlocking with a mask on iPhone devices starting with iPhone 12 and iOS 15.4 or later. Furthermore, it’s designed to work indoors, outdoors, and even in total darkness. An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks. Face ID data, including mathematical representations of a user’s face, is encrypted and available only to the Secure Enclave. This data never leaves the device. It’s not sent to Apple, nor is it included in device backups. The following Face ID data is saved, encrypted only for use by the Secure Enclave, during normal operation:

- The mathematical representations of a user’s face calculated during enrollment
- The mathematical representations of a user’s face calculated during some unlock attempts if Face ID deems them useful to augment future matching

Face images captured during normal operation aren’t saved but are instead immediately discarded after the mathematical representation is calculated for either enrollment in Face ID or comparison with enrolled Face ID data.

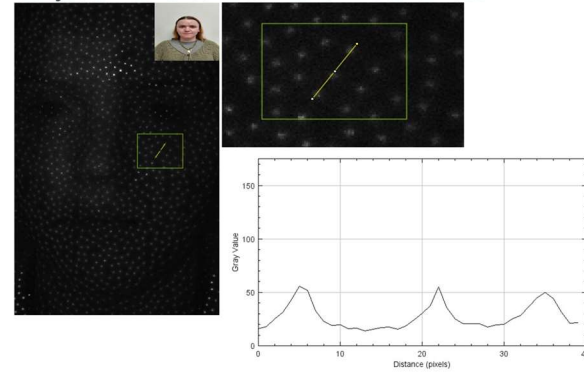
Exhibit 35, at 29.

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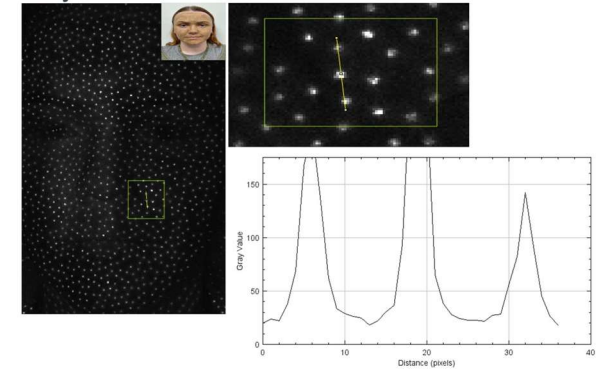
	Advanced technologies The technology that enables Face ID is some of the most advanced hardware and software that we’ve ever created. The TrueDepth camera captures accurate face data by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face. A portion of your device's neural engine — protected within the Secure Enclave — transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data. Exhibit 36. Face ID matches against depth information, which isn’t found in print or 2D digital photographs. It’s designed to protect against spoofing by masks or other techniques through the use of sophisticated anti-spoofing neural networks. Face ID is even attention-aware, and Face ID with a mask will always confirm attention. Face ID recognizes if your eyes are open and your attention is directed towards the device. This makes it more difficult for someone to unlock your device without your knowledge (such as when you are sleeping). Exhibit 36.
[1e] wherein the processor is configured for determining the material property m by evaluating the material feature ϕ_2m.	The Accused Apple Products comprise a processor that is configured for determining the material property m by evaluating the material feature ϕ_2m.

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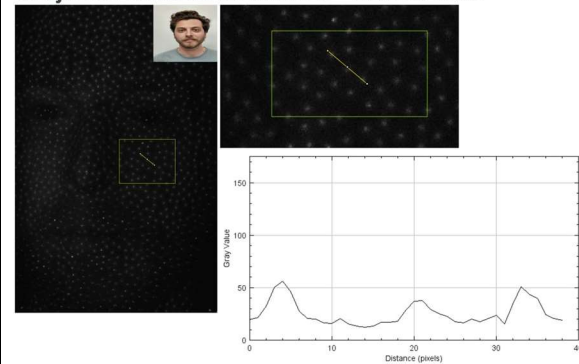
Subject01 Covered area: none – natural Face



Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face

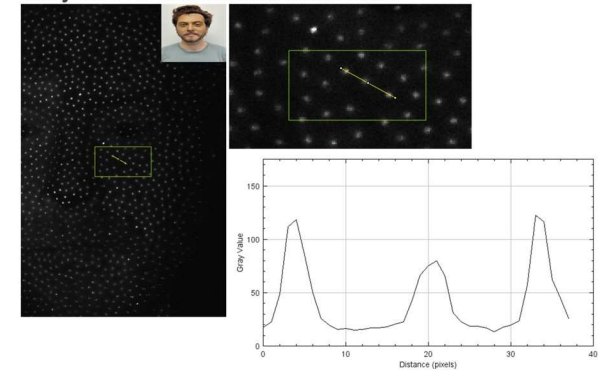


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Ex. 22 - Infringement Contention Claim Chart
U.S. Patent No. 12,332,352 (“the ’352 patent”)

Advanced technologies

The technology that enables Face ID is some of the most advanced hardware and software that we’ve ever created. The TrueDepth camera captures accurate face data by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face. A portion of your device's neural engine — protected within the Secure Enclave — transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data.

Exhibit 36.

Face ID matches against depth information, which isn’t found in print or 2D digital photographs. It’s designed to protect against spoofing by masks or other techniques through the use of sophisticated anti-spoofing neural networks. Face ID is even attention-aware, and Face ID with a mask will always confirm attention. Face ID recognizes if your eyes are open and your attention is directed towards the device. This makes it more difficult for someone to unlock your device without your knowledge (such as when you are sleeping).

Exhibit 36.